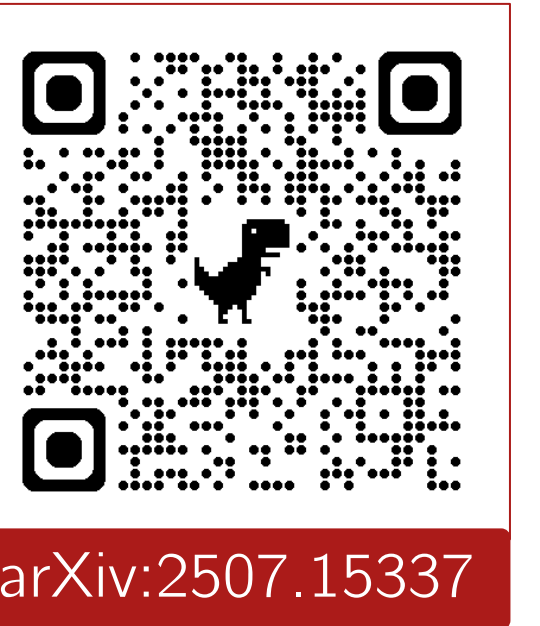




Reasoning Models are Test Exploiters: Rethinking Multiple Choice

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Free-Text

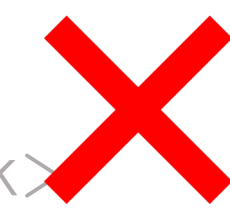


User

A machine starts with 7 tokens. Each minute, the number of tokens triples and then 4 tokens are removed. How many tokens are there after 3 minutes?



<think> Hmm, this requires thought </think>
After some thinking the answer is 51



Multiple-Choice



Benchmark designer

A machine starts with 7 tokens...
A. 133
B. 137
C. 141
D. 149



<think> Hmm, this requires thought </think>
After some thinking the answer is B.

Magnitude anchoring

Backsolving

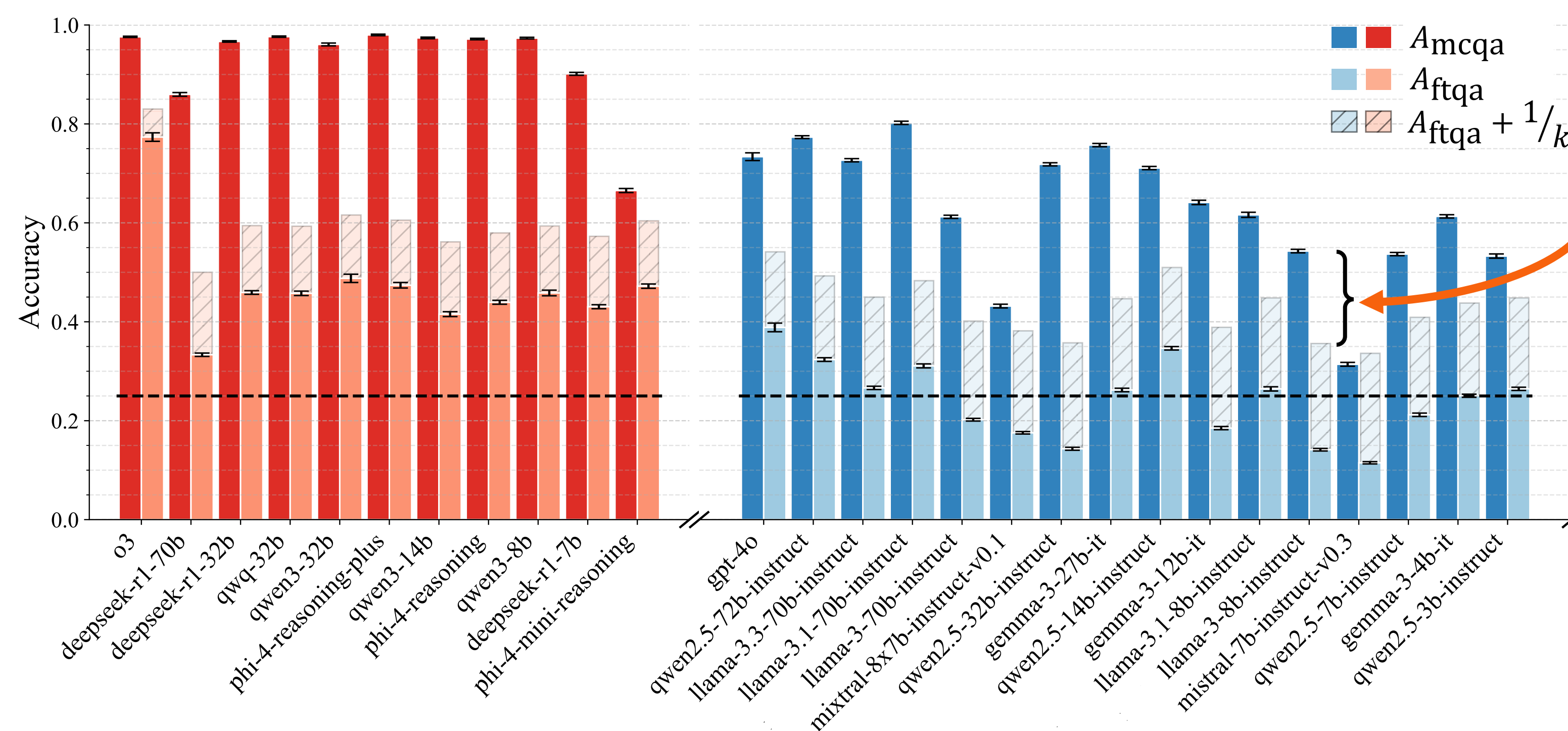


- MCQA is popular because it is **easy to parse**
 - Comprises 75% of tasks in OpenAI's blog on "Learning to Reason with LLMs", in 66% of tasks in Llama 3.1's announcement, and 32% of tasks in HELM
- But "reasoning" allows models to **exploit** information in the options—so reported accuracy can partly measure test-taking skill, not just task competence

Can we calibrate what MCQA is actually measuring by quantifying **exploitation**?

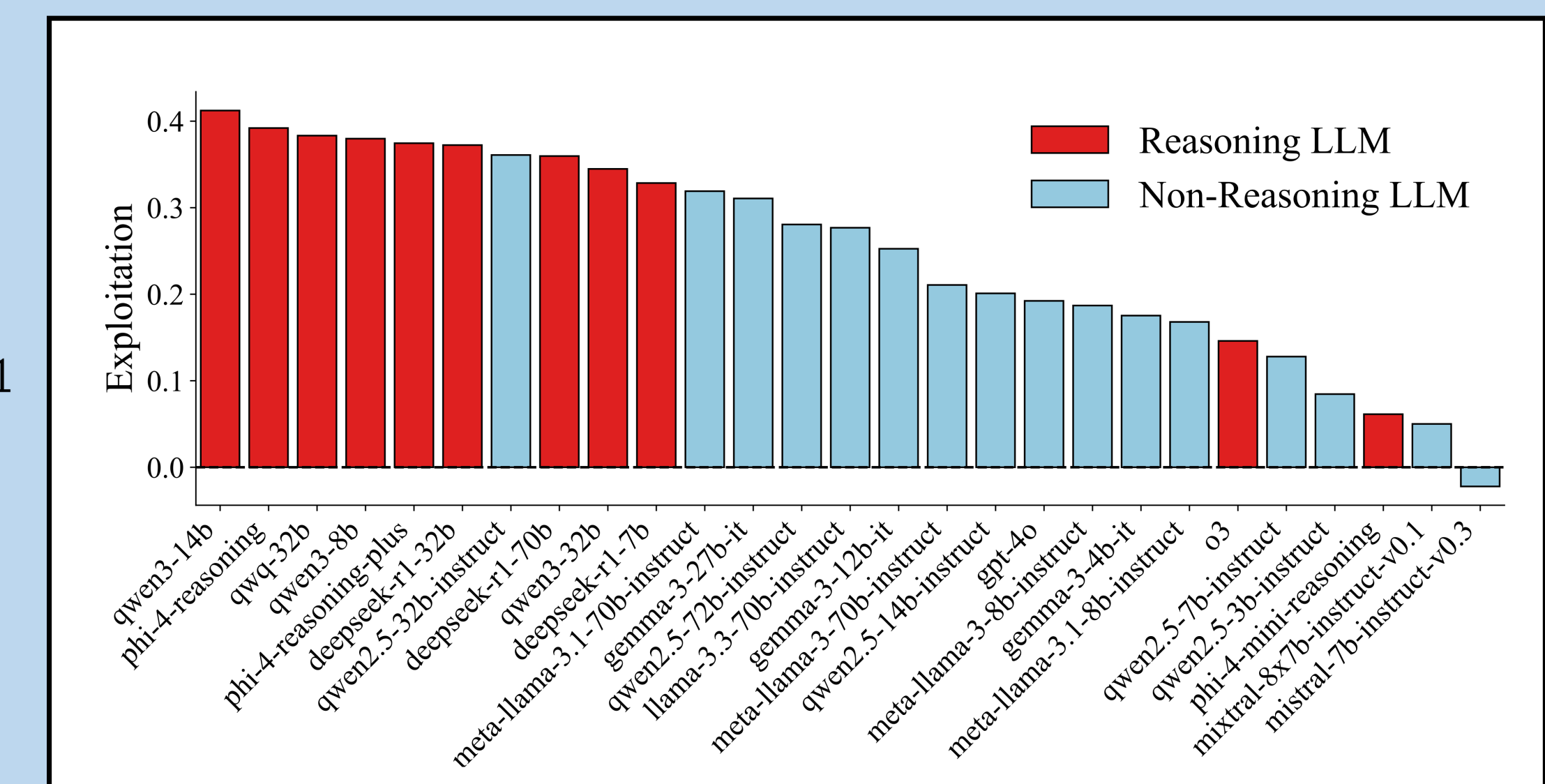
What *exactly* is exploitation?

The excess accuracy (additive) a model achieves from having access to options.



$$E = \left(A_{mcqa} - \frac{1}{k} \right) \cdot \left(\frac{k}{k-1} \right) - A_{ftqa}$$

Diagram illustrating the exploitation index calculation. It shows a stack of bars representing accuracy components. The top bar is labeled $\frac{1}{k}$. Below it, a bar is labeled 1. To the right, an orange circle is labeled 1, followed by $\neq$, then a green circle labeled 0, and finally a green bar labeled 1.



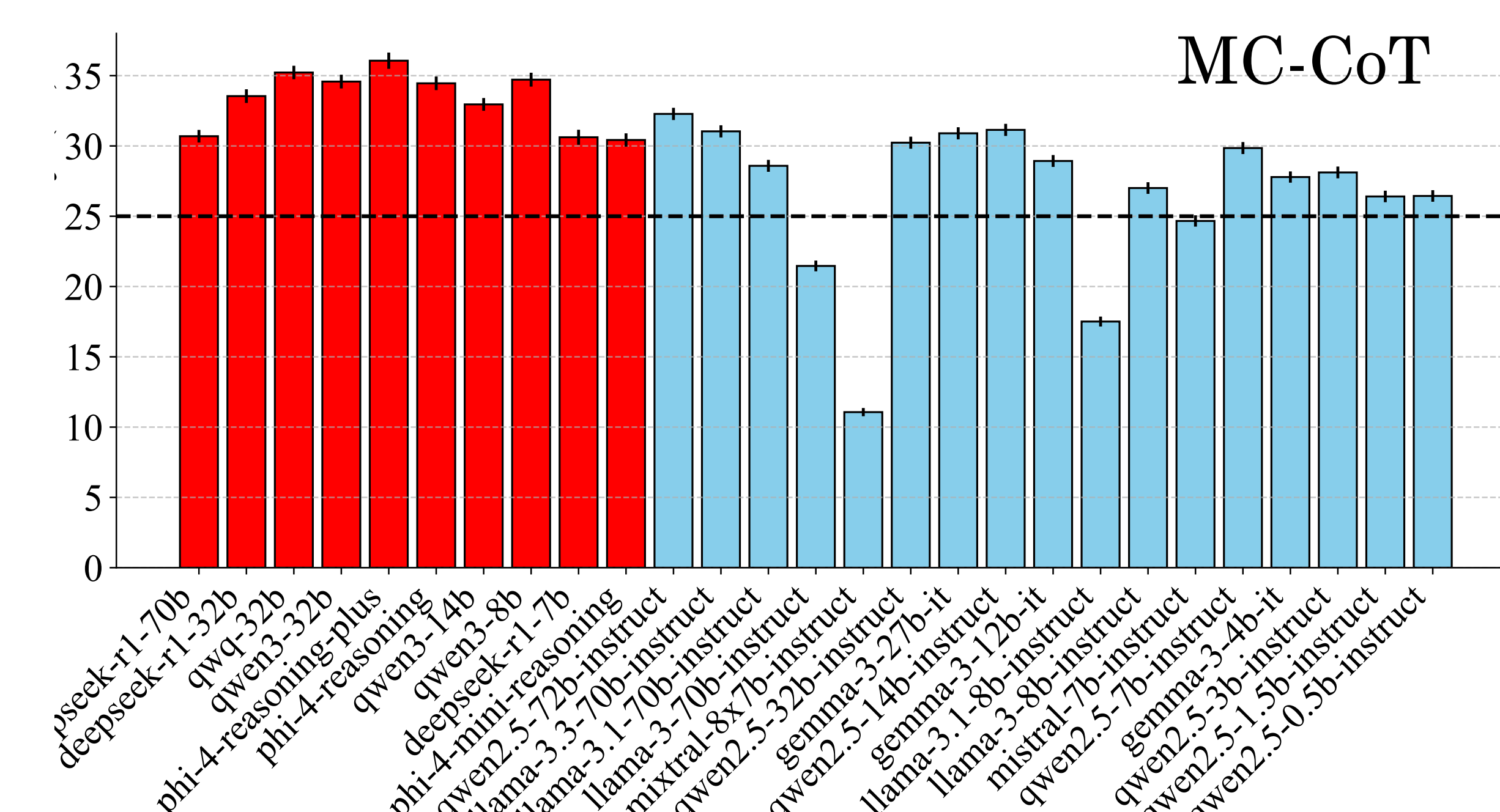
Most reasoning models have an **exploitation index** of over 30%!

What are the *pathways* of exploitation?

Only about 23% of the gap is due to selecting the closest answer.

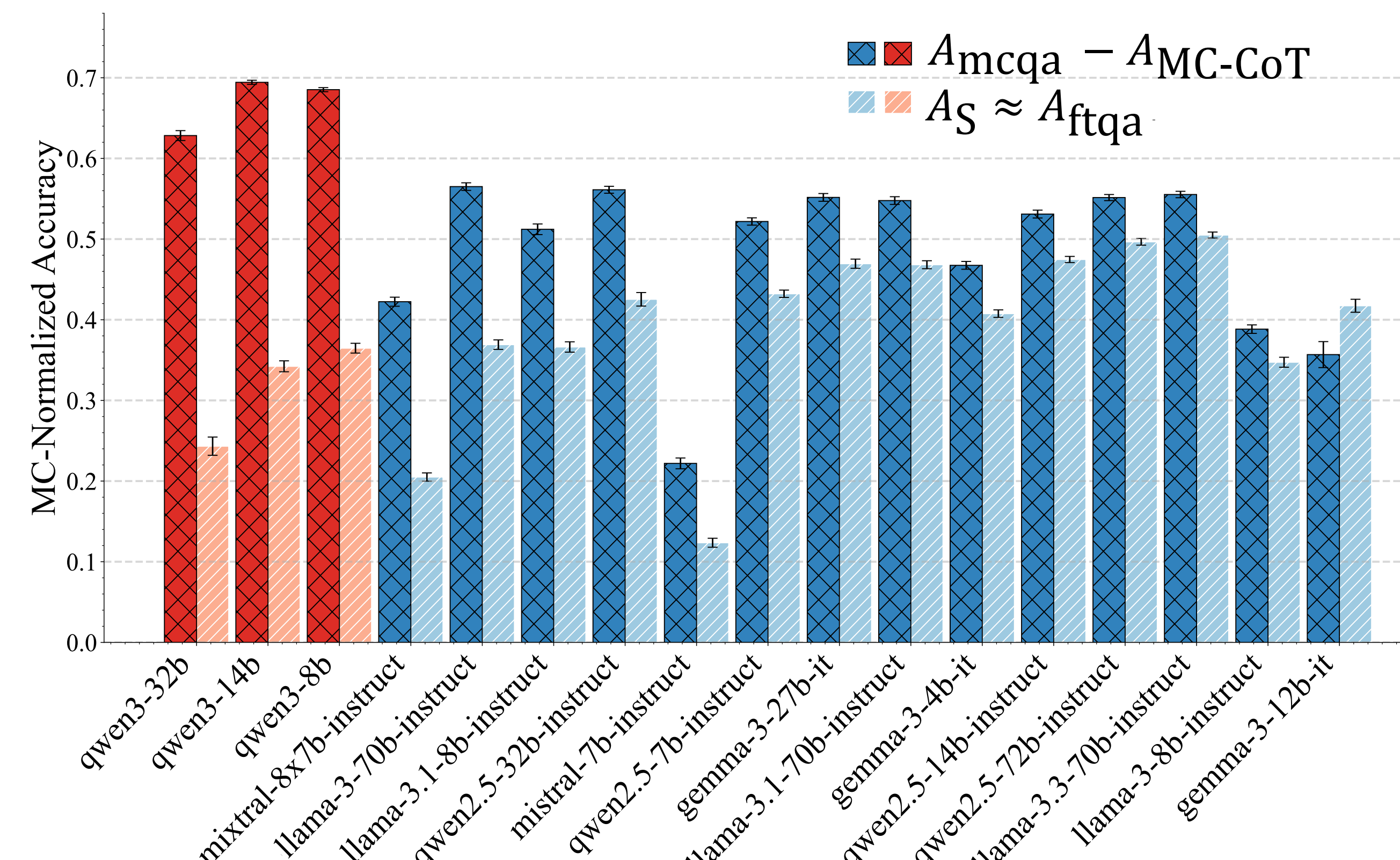
MC-Only Exploitation

Given only access to the options, the performance above random guessing.



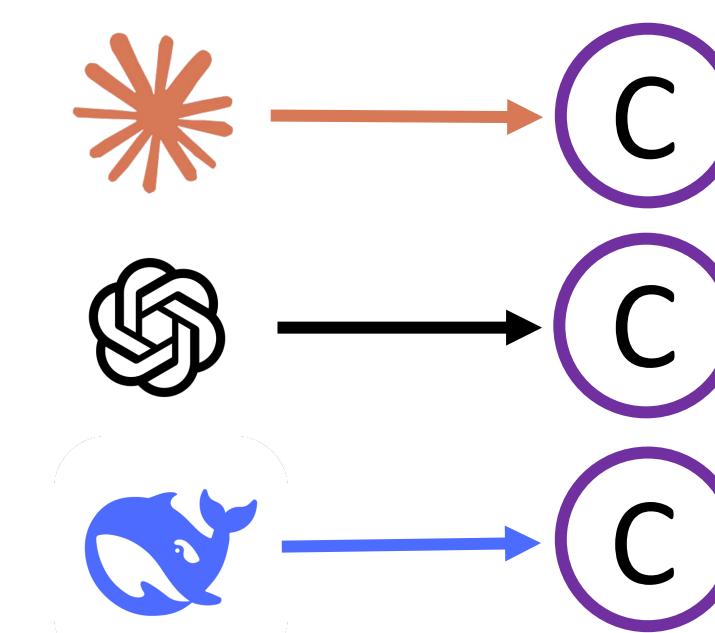
QMC-based Exploitation

The residual exploitation after normalizing for MC-Only exploitation.

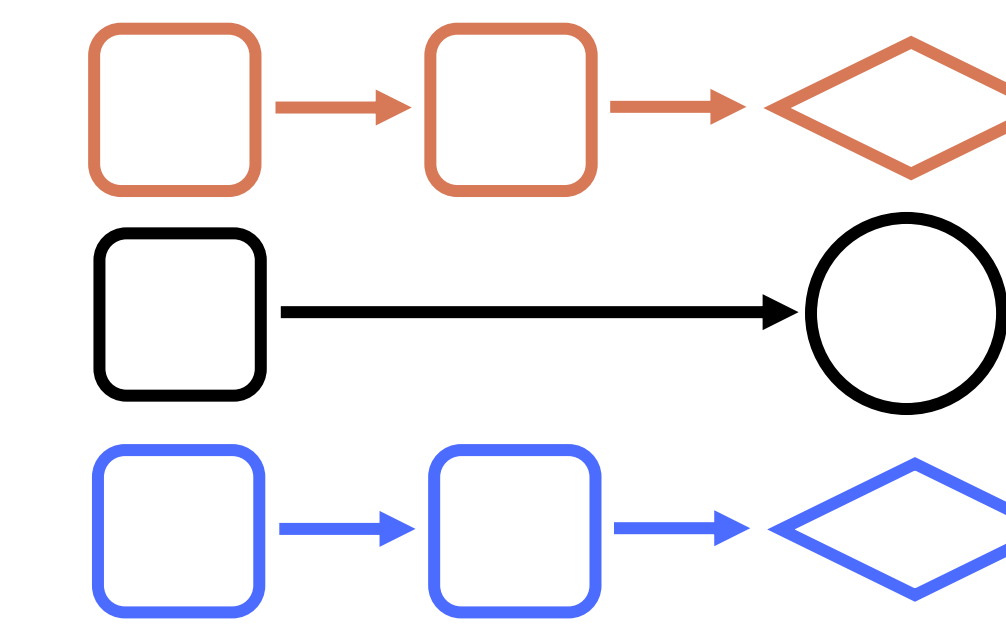


MCQA hides *real* failure mode

Same wrong letter



Different reasoning



We *instead* asked models to output their reasoning in code to diagnose when models have **similar reasoning errors**.

- In over 50% of model pairs, we find higher similarity than MCQA, esp. between Qwen and DeepSeek.
- On 18 problems, we even observed groups dominated by a single family (e.g., a reasoning error type only shared by Gemma models), indicating a systematic failure.

